

# IDEATION PHASE: BRAINSTORM & IDEA PRIORITIZATION

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**Project Name:** Dog Breed Identification using Transfer Learning

## STEP-1: TEAM GATHERING AND PROBLEM SELECTION

Brainstorming provides an open environment for creative thinking to lead to effective problem-solving. For this project, the team focused on identifying 120 different dog breeds with high accuracy.

- **Identified Problem Area:** Manual dog breed identification is time-consuming, requires expert knowledge, and is prone to human error due to fine-grained visual similarities between breeds.
- **Finalized Problem Statement:** Current manual methods are inefficient for large-scale classification. There is a need for an automated AI system capable of identifying 120 breeds from single images (JPG/PNG) using deep learning techniques.

## STEP-2: BRAINSTORMING, IDEA LISTING AND GROUPING

During this phase, we prioritized volume over value to generate diverse ideas for the classification system.

| Generated Ideas                        | Idea Grouping                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------|
| • Manual Feature Extraction (SIFT/ORB) | • <b>Traditional Computer Vision:</b> Using hand-crafted features and classical algorithms. |
| • Support Vector Machines (SVM)        | • <b>Machine Learning:</b> Classification using traditional ML algorithms.                  |
| • Building a Custom CNN from scratch   | • <b>Deep Learning:</b> End-to-end CNN-based classification.                                |

|                                                                                          |                                                                                                                         |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>MobileNetV2 Transfer Learning</b></li> </ul> | <ul style="list-style-type: none"> <li>• <b>Transfer Learning:</b> Leveraging pre-trained CNN architectures.</li> </ul> |
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### STEP-3: IDEA PRIORITIZATION

To determine the best path forward, the team evaluated the ideas based on specific criteria to ensure the solution was both important and feasible.

#### Evaluation Criteria:

- **Accuracy Potential:** Ability to reach >60% validation accuracy.
- **Computational Efficiency:** Fast inference speed (<100ms) on standard hardware.
- **Implementation Complexity:** Ease of building a production-ready system.
- **Scalability:** Capability to handle 120 distinct classes.

#### Selected Approach (Chosen Solution):

The team selected **Transfer Learning using MobileNetV2** .

- **Why:** It provides higher classification accuracy, faster convergence (training in ~30-60 minutes), and efficient feature extraction while reducing overall training complexity compared to building a model from scratch.
- **Outcome:** Achieved a validation accuracy of **67.26%** with a lightweight model size suitable for web deployment.